V3 empirical study · Instruction

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

Shared market sample and filters (not model-specific)

- For each company × regime, one listed-call sample is drawn once and reused by all six models: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S, K, C; (2) no-arbitrage C ≥ max(0, S−K); (3) 7 ≤ DTE ≤ 60; (4) |S/K − 1| ≤ 10%; (5) premium ≥ 0.05, valid bid–ask with relative spread ≤ 50% when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: (κ, θ, ξ, ρ, v0) are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces μ → r.
- Euler ordering: the stock increment over [t, t+Δt] uses the current variance v_t; v_{t+Δt} is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud (n_paths = 2000, seed 42, n_steps = DTE, Δt = 1/252). Paths are not averaged before stopping.

Metrics

- Primary: RMSE% = 100 × √ mean(((C_model − C_mkt) / C_mkt)²). Ranking uses this number only.
- Secondary: MAE = mean |C_model − C_mkt| (dollars). Bias $ = mean(C_model − C_mkt) (dollars). Bias% = 100 × mean((C_model − C_mkt)/C_mkt). Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all six models)

Regime	Window	SPY n	AAPL n	MSFT n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	|S/K−1|≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655

*Input is the processed call panel restricted to (regime end − 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date ≤ t. LSM prices the Monday contract using the latest calibration row with date ≤ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid–ask mid when the source mark is $0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All six models still share that exact sample.

Heston / Heston–Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity × moneyness strata). GBM / Merton / GARCH / GARCH–Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=2000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	58.45	1.4530	+0.3503	+23.87	26.3%	6
GARCH	55.06	1.6866	+1.6498	+44.69	27.9%	4
Heston	33.94	1.1172	+0.6042	+21.77	29.1%	2
Merton	53.28	1.3630	+0.2076	+19.26	24.2%	3
GARCH–Merton	58.05	1.8574	+1.8187	+48.43	25.2%	5
Heston–Merton	**33.19**	**1.0903**	**+0.6216**	**+21.38**	**25.0%**	**1**

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	39.54	0.7003	+0.5086	+26.46	29.2%	2
GARCH	**39.38**	**0.7038**	**+0.5049**	**+26.25**	**27.7%**	**1**
Heston	53.15	1.0265	+0.9581	+42.98	30.6%	5
Merton	39.94	0.7185	+0.4968	+26.21	26.6%	3
GARCH–Merton	44.27	0.8195	+0.6709	+32.98	28.0%	4
Heston–Merton	68.38	1.3109	+1.2509	+55.17	29.0%	6

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	51.52	1.7068	+1.2167	+34.33	24.7%	2
GARCH	69.64	2.1225	+1.7298	+45.00	30.2%	3
Heston	74.11	2.3692	+2.1369	+55.34	25.0%	4
Merton	**49.57**	**1.6637**	**+1.1397**	**+32.37**	**22.3%**	**1**
GARCH–Merton	76.63	2.4045	+2.1179	+54.42	28.4%	5
Heston–Merton	86.87	2.8636	+2.6435	+66.91	21.5%	6

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	66.95	3.6931	+1.7288	+45.87	26.9%	5
GARCH	**59.32**	**3.0718**	**+2.3450**	**+37.04**	**38.1%**	**1**
Heston	66.39	3.8056	+2.6632	+50.76	30.2%	4
Merton	62.95	3.5023	+1.4543	+41.75	21.6%	2
GARCH–Merton	63.13	3.3311	+2.5402	+43.61	37.0%	3
Heston–Merton	67.60	3.5696	+2.4566	+48.77	18.6%	6

RMSE% is $100\times\sqrt{\text{mean}(((\text{model}-\text{market})/\text{market})^2)}$. MAE and Bias $ are in option-premium dollars. Bias% is $100\times\text{mean}((\text{model}-\text{market})/\text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 5 · Table block · AAPL · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=2000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	61.13	2.5357	+1.5246	+33.58	24.8%	5
GARCH	53.39	2.8266	+2.6794	+43.19	23.7%	3
Heston	**32.43**	**1.6546**	**+1.1227**	**+21.96**	**27.1%**	**1**
Merton	57.68	2.4293	+1.3903	+30.98	23.2%	4
GARCH–Merton	61.41	3.4149	+3.3207	+52.81	22.9%	6
Heston–Merton	34.23	1.6730	+1.1881	+23.23	21.9%	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	52.06	3.6666	+3.6326	+40.10	28.6%	5
GARCH	47.66	3.2942	+3.1762	+31.21	27.1%	4
Heston	38.62	2.5601	+2.5591	+30.71	26.3%	2
Merton	47.54	3.3845	+3.3397	+35.82	23.4%	3
GARCH–Merton	68.15	5.1518	+5.1479	+55.83	22.5%	6
Heston–Merton	**32.93**	**2.0632**	**+2.0407**	**+22.48**	**21.0%**	**1**

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	34.01	1.7715	+1.1125	+20.36	26.3%	2
GARCH	48.92	2.0791	+1.6369	+30.32	26.3%	5
Heston	45.30	2.3807	+2.3093	+38.77	28.5%	4
Merton	**32.33**	**1.6676**	**+1.0112**	**+18.85**	**23.8%**	**1**
GARCH–Merton	60.90	2.7654	+2.5372	+44.94	24.6%	6
Heston–Merton	44.15	2.2146	+2.0747	+35.15	24.1%	3

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	47.86	4.2247	+2.6177	+32.64	27.8%	4
GARCH	63.11	5.0239	+4.4985	+44.33	34.0%	5
Heston	42.43	3.7882	+3.0597	+33.78	29.9%	2
Merton	46.09	3.8303	+2.1912	+29.59	23.6%	3
GARCH–Merton	72.22	6.2435	+5.7535	+58.33	27.8%	6
Heston–Merton	**41.64**	**3.5324**	**+2.5584**	**+30.70**	**21.5%**	**1**

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths e

Page 6 · Table block · MSFT · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=2000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	46.75	0.3814	+0.1096	+20.10	27.0%	4
GARCH	72.50	0.5968	+0.5538	+55.64	37.3%	5
Heston	**28.27**	**0.2618**	**+0.1014**	**+13.94**	**29.9%**	**1**
Merton	44.85	0.3804	+0.0927	+18.17	23.1%	3
GARCH–Merton	85.11	0.7656	+0.7296	+70.93	32.0%	6
Heston–Merton	28.33	0.2652	+0.1155	+15.14	24.2%	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	64.39	0.4456	+0.4456	+55.00	27.9%	4
GARCH	94.35	0.6883	+0.6883	+82.46	28.2%	5
Heston	32.71	0.1999	+0.1659	+22.40	27.2%	2
Merton	58.34	0.3957	+0.3880	+48.65	21.5%	3
GARCH–Merton	116.07	0.8814	+0.8814	+104.79	27.3%	6
Heston–Merton	**28.90**	**0.1793**	**+0.1345**	**+18.78**	**15.9%**	**1**

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	43.83	1.1281	+0.8619	+29.36	26.1%	4
GARCH	44.65	1.1166	+0.9640	+32.10	26.2%	5
Heston	37.65	0.9636	+0.8374	+26.73	28.7%	2
Merton	42.00	1.0879	+0.8015	+27.38	23.4%	3
GARCH–Merton	54.96	1.4249	+1.3549	+44.02	23.5%	6
Heston–Merton	**34.59**	**0.8448**	**+0.6921**	**+21.89**	**21.0%**	**1**

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	47.79	2.1217	+1.1208	+30.05	27.1%	4
GARCH	66.83	2.6466	+2.1894	+39.01	34.7%	5
Heston	41.31	2.1263	+1.5877	+31.36	28.1%	2
Merton	46.87	2.0173	+0.9824	+28.29	22.6%	3
GARCH–Merton	72.53	3.0574	+2.4901	+45.89	34.0%	6
Heston–Merton	**35.15**	**1.7877**	**+1.1299**	**+21.12**	**22.3%**	**1**

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths e

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 12 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 12 cells bes
1	**Heston**	**43.86**	**39.96**	**2**
2	Heston–Merton	44.66	34.87	6
3	Merton	48.45	47.21	2
4	GBM	51.19	49.69	0
5	GARCH	59.57	57.19	2
6	GARCH–Merton	69.45	65.64	0

Table S2 · Wins by company (lowest RMSE% in that company×regi

Model	SPY wins	AAPL wins	MSFT wins	Total
GBM	0	0	0	0
GARCH	2	0	0	2
Heston	0	1	1	2
Merton	1	1	0	2
GARCH–Merton	0	0	0	0
Heston–Merton	**1**	**2**	**3**	**6**

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	55.44	52.00	43.12	54.20	51.19
GARCH	60.32	60.46	54.40	63.08	59.57
Heston	**31.55**	**41.49**	**52.35**	**50.04**	**43.86**
Merton	51.94	48.61	41.30	51.97	48.45
GARCH–Merton	68.19	76.16	64.16	69.30	69.45
Heston–Merton	31.91	43.40	55.20	48.13	44.66

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Page 8 · Summary by company · mean RMSE% across four regimes

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes bes
GBM	54.12	54.99	0
GARCH	55.85	57.19	2
Heston	56.90	59.77	0
Merton	**51.43**	**51.42**	**1**
GARCH–Mertor	60.52	60.59	0
Heston–Mertor	64.01	67.99	1

Table S4.2 · AAPL

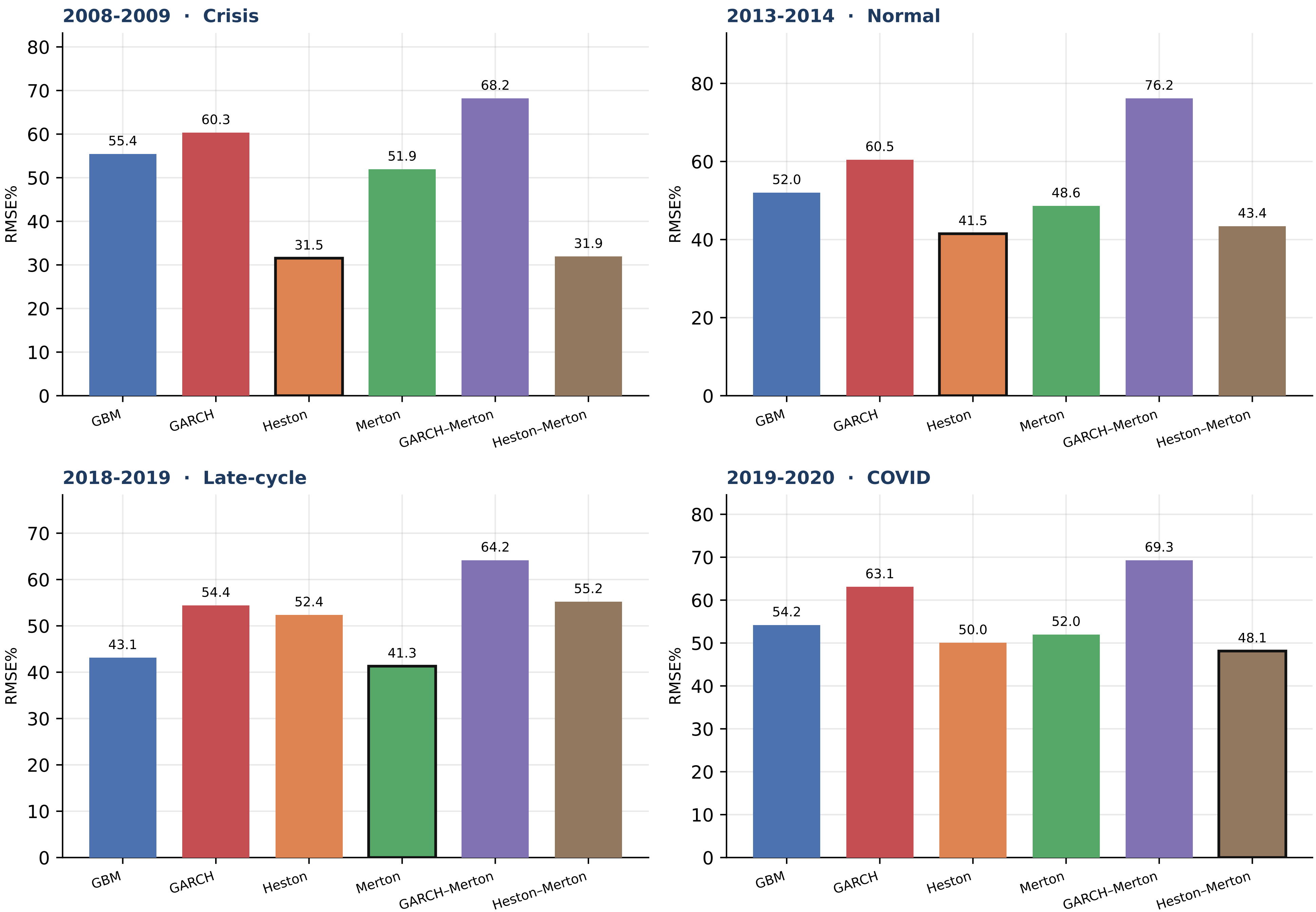
Model	Mean RMSE%	Median	# regimes bes
GBM	48.76	49.96	0
GARCH	53.27	51.15	0
Heston	39.69	40.52	1
Merton	45.91	46.81	1
GARCH–Mertor	65.67	64.78	0
Heston–Merto	**38.24**	**37.94**	**2**

Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes bes
GBM	50.69	47.27	0
GARCH	69.58	69.66	0
Heston	34.99	35.18	1
Merton	48.02	45.86	0
GARCH–Mertor	82.17	78.82	0
Heston–Merto	**31.74**	**31.74**	**3**

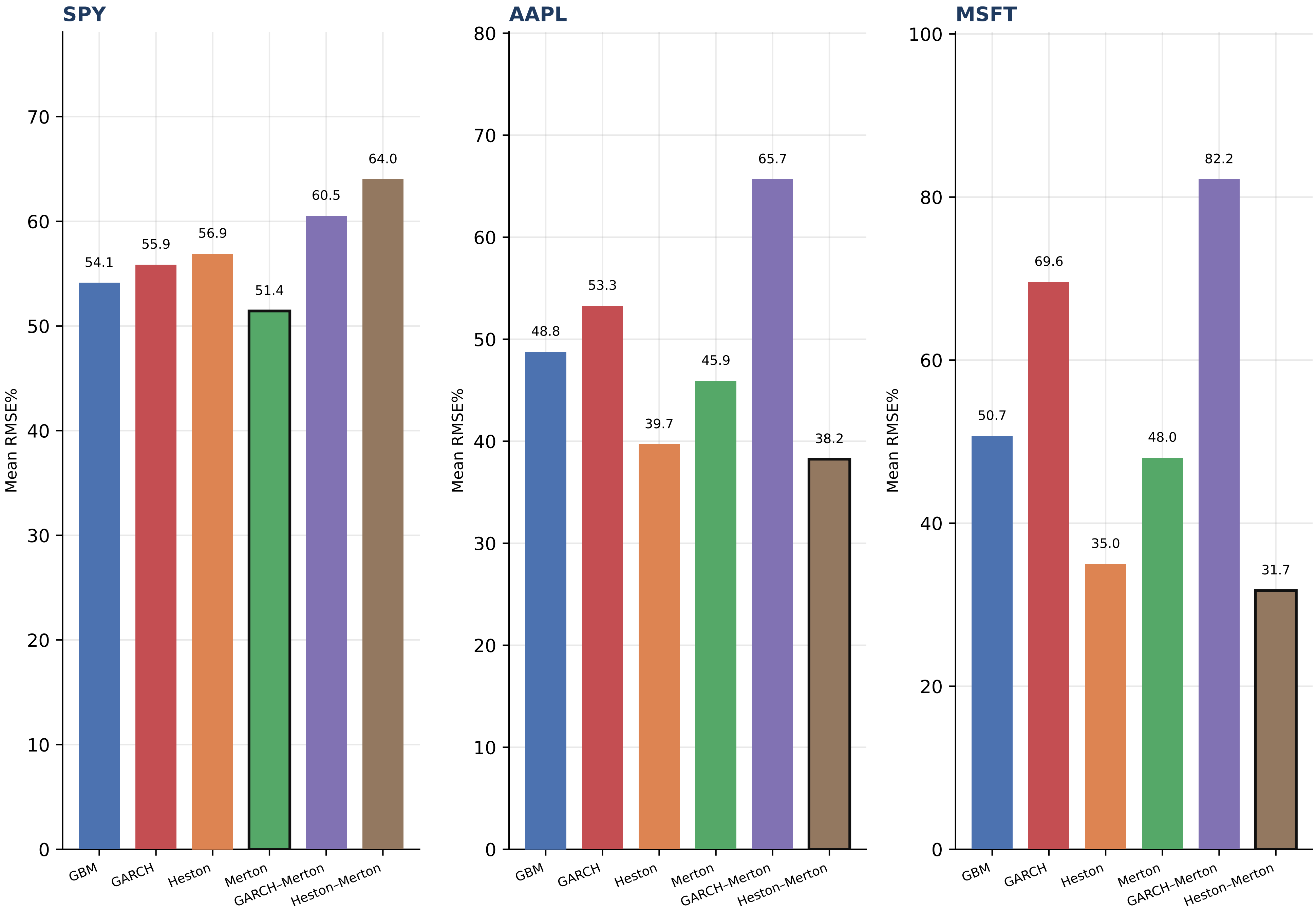
Each column is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.

Page 9 · Figure 1 · RMSE% by model and regime · mean of SPY / AAPL / MSFT



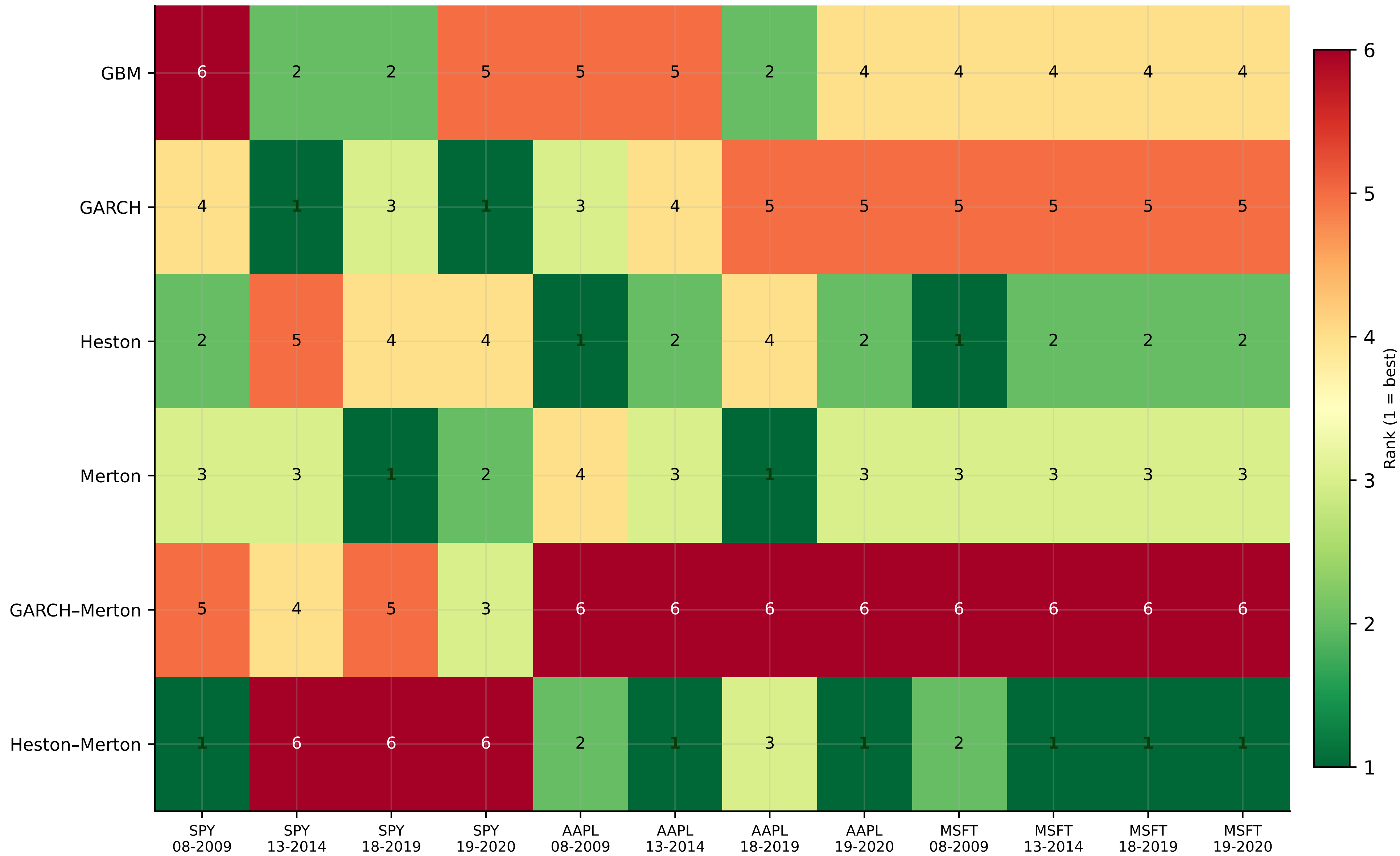
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the three companies.

Figure 2 · RMSE% by model · each company (mean across four regimes)

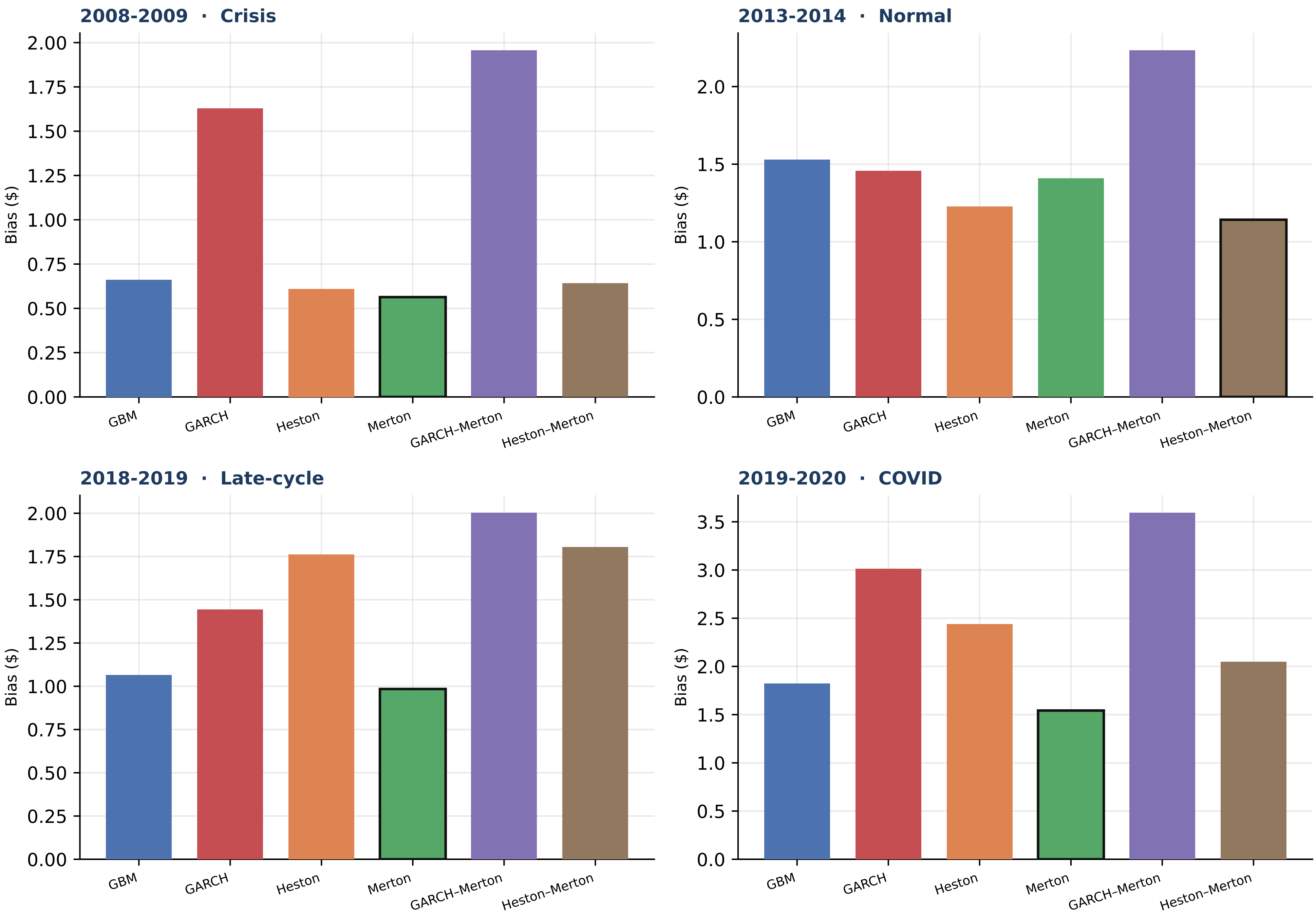


Outlined bar = lowest mean RMSE% for that company.

Page 11 · Figure 3 · Model rank by company × regime (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Page 13 · Final analysis

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 12 company×regime cells, Heston is the best overall model (mean RMSE% = 43.86; 2 of 12 cells).
- Second overall is Heston–Merton (mean RMSE% = 44.66; 6 cells).
- The ranking is not stable. Best-in-cell models across the 12 tables: GARCH, Heston, Heston–Merton, Merton.
- SPY: lowest mean RMSE% is Merton (51.43); wins 1 of 4 regimes.
- AAPL: lowest mean RMSE% is Heston–Merton (38.24); wins 2 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston–Merton (31.74); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston (31.55).
- 2013-2014 (Normal): best mean RMSE% is Heston (41.49).
- 2018-2019 (Late-cycle): best mean RMSE% is Merton (41.30).
- 2019-2020 (COVID): best mean RMSE% is Heston–Merton (48.13).
- No-jump models as a group have lower mean RMSE% (51.54) than the three jump models (54.19).
- For the overall winner (Heston), crisis RMSE% is 31.55 vs 41.49 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 2,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1…MSFT-4 uses exactly the same option contracts for all six models. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias $, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.